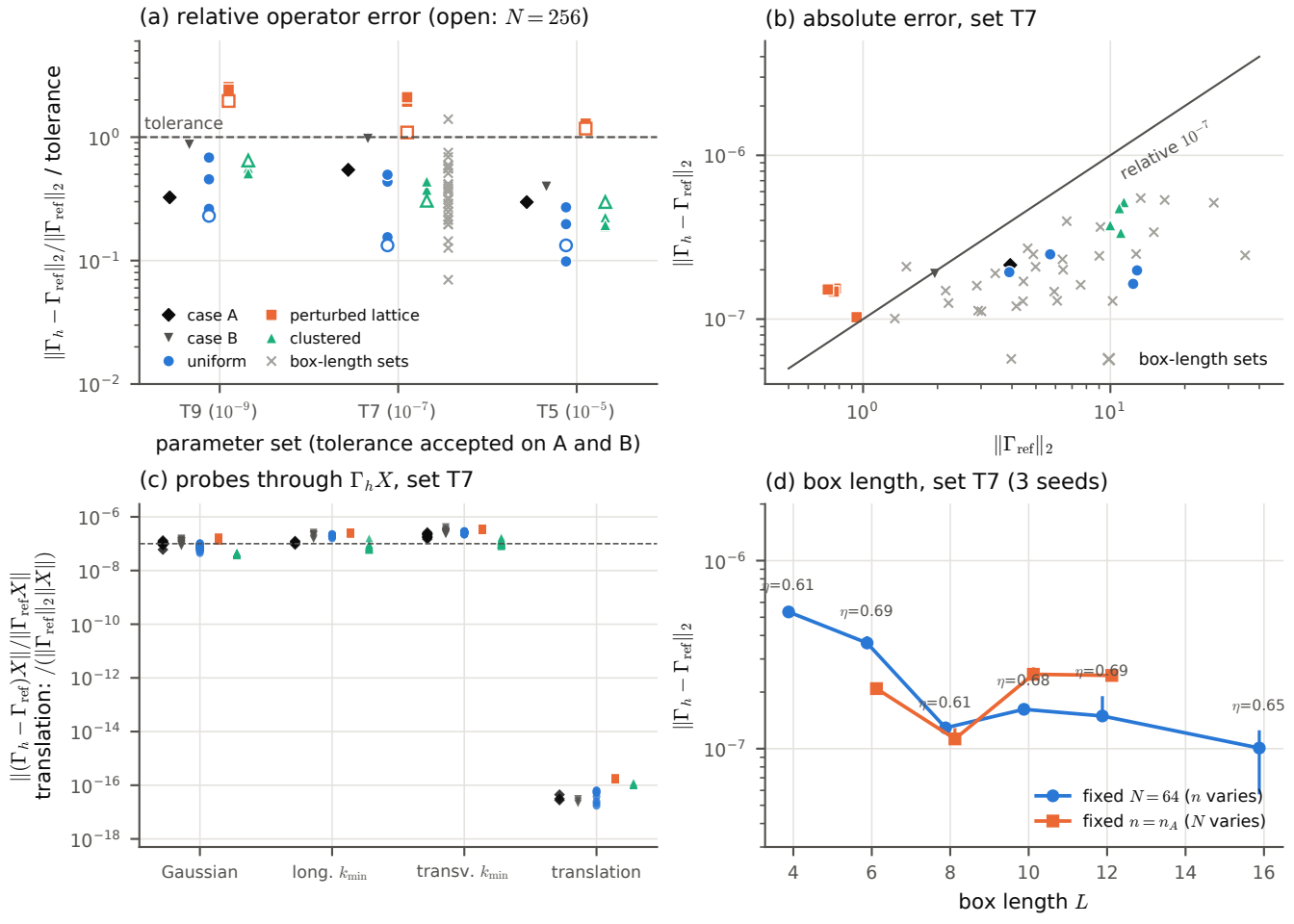
(a) relative operator error (open: $N = 256$)
$\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$ / tolerance
tolerance
10^0
10^{-1}
10^{-2}
case A
case B
uniform
perturbed lattice
clustered
box-length sets
T9 (10^{-9})
T7 (10^{-7})
T5 (10^{-5})
parameter set (tolerance accepted on A and B)
(b) absolute error, set T7
$\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2$
10^{-6}
10^{-7}
relative 10^{-7}
box-length sets
10^0
10^1
$\|\Gamma_{\mathrm{ref}}\|_2$
(c) probes through $\Gamma_h X$, set T7
$\|(\Gamma_h - \Gamma_{\mathrm{ref}})X\| / \|\Gamma_{\mathrm{ref}} X\|$
translation: $/(\|\Gamma_{\mathrm{ref}}\|_2 \|X\|)$
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
Gaussian
long. $k_{\min}$
transv. $k_{\min}$
translation
(d) box length, set T7 (3 seeds)
$\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2$
10^{-6}
10^{-7}
η=0.61
η=0.69
η=0.61
η=0.68
η=0.69
η=0.65
fixed $N = 64$ (n varies)
fixed $n = n_A$ (N varies)
4
6
8
10
12
14
16
box length L